

2024



Solink Reduces Theft by Analyzing 1 Billion Transactions per Month on AWS

Learn how Solink’s security solutions increase businesses’ profits by 2 percent by controlling and reducing shrink on AWS.

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3–5x average return
on investment for businesses

1 billion transactions
analyzed per month

Learn more about Solink's security solutions on AWS.

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Overview

When [Solink](#), a security and loss prevention solutions company, set out to change the face of retail by offering a comprehensive physical security solution, it looked to the cloud to provide the scalability and availability it needed. The company's product combines video surveillance with transaction data to help retailers assess and identify risk across thousands of locations, access footage of incidents, and drill into the details of every transaction.

Solink chose to build on Amazon Web Services (AWS) and to use artificial intelligence (AI) for video footage analysis. On AWS, Solink is scaling with a lean engineering team while providing reliable, user-friendly software to its customers.

Opportunity | Bringing Loss Prevention and Video Intelligence to Tens of Thousands of Locations Globally

As a company born in the cloud, Solink provides video analysis and AI-powered analytics to retailers who have historically treated security cameras as an insurance policy to manually retrieve footage after an incident takes place. Security cameras capture every second of every interaction, but without continuous monitoring, the sheer volume of data from these on-premises cameras often goes unseen, or worse, the cameras can be offline. Analyzing and bubbling up the most important risk is critical, but human-based surveillance requires hours of review time—a classic “needle in a haystack” problem. “We pair video from surveillance cameras with data, such as point-of-sale systems, to provide context and generate insights,” says Jim Farrell, vice president of sales at Solink. By consolidating the data silos around live security footage, Solink can quickly and accurately identify suspicious or risky activity to protect businesses. Solink manages tens of thousands of sites in more than 40 countries and monitors more than 1 billion transactions per month.

In looking for a cloud infrastructure provider, Solink's main criterion was that it needed a foundation for scalable, reliable, highly available, and secure infrastructure. The company chose AWS and took advantage of [AWS Regions](#), physical locations around the world where AWS clusters its data centers, to go global with its multinational customers. “When we started engineering and architecting our solution, we looked for a cloud provider that would give us the infrastructure to scale without requiring a lot of work,” says Norm Wong, chief technology officer at Solink. The team began designing the cloud offering in 2015 and launched it in 2016. “We used a lot of solutions that AWS provided,” says Wong. “Having high availability across Availability Zones in an AWS Region was especially important to us.” Solink also received cost-optimization support from the AWS team while it built a solution capable of handling hundreds of thousands of cameras with low latency.

Solution | Delivering Three to Five Times Return on Investment for Customers Using AI on AWS

Solink's solution creates a pipeline built on AWS to continuously process live security footage from customers. The pipeline uses a hybrid model that records video at the edge while continually determining what footage needs to be cloud processed. Certain segments of the footage are stored in [Amazon Simple Storage Service](#) (Amazon S3), which provides object storage built to retrieve virtually any amount of data from anywhere. Solink uses many other AWS solutions, including image analysis from [Amazon Rekognition](#), which automates and lowers the cost of image recognition and video analysis with machine learning.

The company has a Video Alarms offering that outperforms legacy "sensor/panel"-based intrusion detection (alarm system). When the system detects suspicious activity in an "armed" state, Solink's computer vision algorithms, built on AWS, determine if there's a person in the environment and if he or she poses a risk, thereby minimizing false alarms. Alongside this feature, Solink uses [Amazon SageMaker](#), a fully managed service that brings together a broad set of tools for high-performance, low-cost machine learning, to continuously train and fine-tune models running in production.

By helping reduce external and internal theft, Solink offers its customers an average in-year return on investment of three to five times. "Our system significantly reduces internal theft and fraud because once you're using Solink, risk monitoring becomes an internal workflow that is on autopilot," says Farrell. "We identify illicit activity for review, turning your security investment from passive to active." Additionally, the ability to save and share video in the cloud reduces legal costs and maintains the chain of custody of evidence for legal cases. "We can reduce shrink by up to 70 percent for a company with Solink," says Farrell.

Solink's customers access their video remotely using a secure relay technology that Solink developed. The relay clusters are located in distributed AWS Regions to minimize latency and avoid firewall "hole-punching"—a risky cybersecurity procedure that traditional on-premises network video recorders and digital video recorders require for remote access. Solink has continued to invest in security in the cloud to bring daily improvements to cybersecurity and privacy controls, and the company completed its SOC 2 Type 2 in 2022.

Solink has built a foundation to evolve features and meet new customer expectations quickly and iteratively. "One of the advantages of the cloud we saw early on was that we could evolve the product with direct feedback from customers much faster than on-premises providers," says Martin Soukup, vice president of engineering at Solink. "We have a continuous integration and delivery development cycle to push hundreds of releases a year. Moreover, we can elastically scale on AWS as demand fluctuates." The broad reach of AWS and the flexibility of AWS services also mean that Solink can provide a solution that is responsive, secure, and modern. "We can meet our customers where they are with technology and security overall," says Farrell. "That gives us a major advantage over legacy systems that package software in the hardware shipped on site."

Outcome | Adopting Cutting-Edge Technologies to Further Improve Loss Prevention

True to its foundation in the cloud, Solink is investing in generative AI solutions such as Solink Sidekick AI, which the company launched in 2023. "Solink has a highly scalable system on AWS, and we will continue to use the scale and distribution of AWS to implement new technologies and deliver more value to our customers," says Soukup.

As it continues to grow, Solink is positioned to help even more business owners gain visibility assets. "By running our solution on AWS, we're providing an enterprise-grade, mission-critical



hassle of building out infrastructure," says Soukup.

About Solink

Solink offers trusted cloud video surveillance systems for businesses of all sizes. Its hardware and software help give visibility to IT, loss prevention, and security teams at tens of thousands of locations in more than 40 countries.

AWS Services Used

Amazon Rekognition

Automate and lower the cost of your image recognition and video analysis with machine learning.

[Learn more »](#)

Amazon SageMaker

Amazon SageMaker is a fully managed service that brings together a broad set of tools to enable high-performance, low-cost machine learning (ML) for any use case.

[Learn more »](#)

Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

[Learn more »](#)

AWS Regions



AWS has the concept of a Region, which is a physical location around the world where we cluster data centers.

[Learn more »](#)

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